Denizen — Scene Building Handbook

Cut, label & place pixel sprites · build rooms and boundaries by hand · no coding required.

Quick Start (the one page that matters)

1 · Get in

1. Open a terminal in the `denizen/` folder → run `npm start`
2. Browser → `http://localhost:8080/` (this is the game)
3. Click the game, press **E** → the **EDITOR** opens. **E** again closes it.

> Nothing loads? The server isn't running — start it first. Keys do nothing while a text box is focused; click empty floor first.

2 · The keys

Key	Does
E	Open / close the editor
X	Save the active tab (furniture layout / walls file)
Double-click a piece	Delete it
D	(Walls) Duplicate selected wall
Delete	(Walls) Delete selected wall
K	(Walls) Cut a wall — make a doorway gap

3 · Build a room (in order)

① **Walls tab** — *Rebuild perimeter box* for an instant room, then **K** to cut doorways. Or pick **Draw wall** and drag: across = top wall, up/down = side wall, diagonal = corner. **Walls are the boundaries.** Turn on **Show Collisions** to see them.

② **Furniture tab** — **Search** a name → **click the item** → **click the floor** to drop it. Keep **Snap: ON**. Leave ~2 tiles of walking space between facing desks.

③ **Save** — press **X** on each tab.

4 · Naming a NEW object

The editor only lists items that have been **named**. To add one:

Art is…	Open this page → do this
LimeZu themed piece	`/tools/limezu-sprite-browser.html` — type a name → Save
Modern Office desk/chair	`/sprite-cutter.html` then `/sprite-labeler.html` — cut a box, name + type → Save
Floor / wall tile	`/tools/tile-inspector.html` — click tile → copy frame #

5 · Copy a finished room

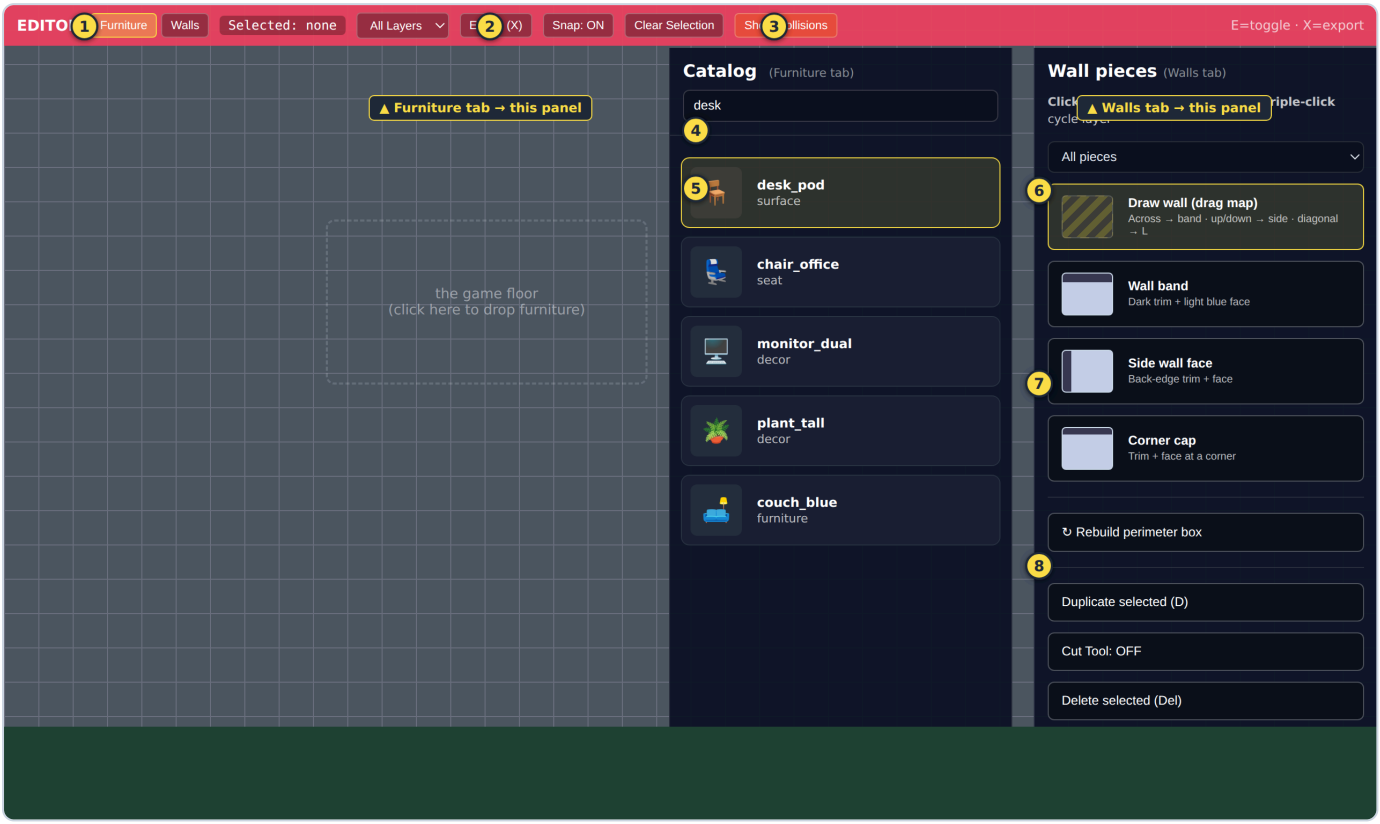
In `assets/limezu/examples/` : `*_layer1.png` =walls (build first), `*_layer2.png` =furniture, `*_preview.png` =the goal. Source packs at **limezu.itch.io**.

6 · Hand back

After **X**: `furniture_catalog_openplan_edited.json`, `openplan_walls_edited.json`, `data/limezu-labels.json`. That's the whole job. ✅

The Editor Panel — every button

Opens when you press **E** in the game. Both side panels are shown together below; in the game you see one at a time — the **Furniture** tab shows the Catalog panel, the **Walls** tab shows the Wall pieces panel.



Reproduced 1:1 from the editor markup in office-scene.js — the buttons match what you'll see.

#	Control	What it does
1	Furniture / Walls tabs	Switch what you're editing. The active tab decides which side panel shows and what **X** exports.
2	Export (X)	Save the active tab. Furniture → layout saved to the live game + a backup .json downloads. Walls → `openplan_walls_edited.json` downloads.
3	Show Collisions	Toggle the invisible boundaries. **Red** = solid things that block movement. **Blue** = squares no one can walk on. Your check that walls/paths are right.
4	Search box	(Furniture) Type part of a name to filter the catalog. Only **named** items appear.
5	Catalog item	(Furniture) Click one (highlights yellow), then click the floor to drop it. Double-click a placed piece to delete it.
6	Draw wall	(Walls) Drag on the floor: across → top "band" wall, up/down → side wall, diagonal → an L-corner.
7	Wall piece buttons	(Walls) Drop one specific piece: Wall band, Side wall face, Corner cap, T-junction.
8	Wall actions	(Walls) Rebuild perimeter box = instant clean room. Duplicate (D), Cut Tool (K) = doorway, Delete (Del), Export walls (X).

Also on the toolbar: Selected shows the clicked piece's id · **All Layers** dropdown dims everything except one type · **Snap: ON** keeps things grid-aligned (leave on) · **Clear Selection** deselects.

The Big Picture

A "scene" is built from three things, in this order: **1) the shell** (floor + 3D walls), **2) the boundaries** (invisible walls that stop people walking through stuff), **3) the furniture** (placed on the floor).

Before furniture can go in, the game has to **know what each sprite is**. The art packs ship as big sheets with **no names** — `Singles_42` is just "the 42nd picture the artist exported," not "blue chair." So there are two jobs, and they feed each other:

Job	What you do	Where
A. Catalog	Cut each sprite out, give it a name, save it to the catalog	Standalone tool pages
B. Build	Drag the named sprites into a room, draw walls, set boundaries	In-game editor (press **E**)

> **Golden rule:** never draw cut-boxes on a LimeZu *theme* sheet (`themes/japanese.png` etc.) — those are lookup pictures only. The real game-ready pieces are the **Singles** (one file each, already cut); you only **name** them.

Which tool for which art?

Working with…	It's a…	Use
LimeZu themed object (Japanese, gym, kitchen, lounge, library, conference)	"Single" (one PNG)	Name it in `limezu-sprite-browser.html`
Modern Office desk / chair / monitor (HQ look)	Piece on a big sheet	Cut: `sprite-cutter.html` · Name: `sprite-labeler.html`
The floor or walls of a room (empty shell)	Tile on a grid	Frame # in `tile-inspector.html`
Placing named things / drawing boundaries	—	Open the game, press **E**

The tool pages (addresses)

Tool	Address	For
Sprite Cutter	`/sprite-cutter.html`	Draw a box around one object on a sheet
Sprite Labeler	`/sprite-labeler.html`	Name + type each Modern Office cut
LimeZu Sprite Browser	`/tools/limezu-sprite-browser.html`	Name LimeZu Singles (display name, footprint, depth)
Tile Inspector	`/tools/tile-inspector.html`	Pick floor/wall tile frame numbers
Room Shell Preview	`/tools/room-preview.html`	Preview an empty room (floor + walls)
Single Preview	`/tools/limezu-single-preview.html`	Test that one sprite loads correctly
Verify Sprites	`/verify-sprites.html`	Final QA — every catalog entry cropped at once
Catalog Explorer	`/catalog-explorer.html`	System overview + naming schema

Copy the Artist's Rooms

Don't invent layouts. LimeZu shipped finished example rooms in `assets/limezu/examples/` — use them as the picture you're copying.

Reference room	Files
Generic interior	generic_preview / layer1 / layer2
Gym	gym_preview / layer1 / layer2
Japanese house	japanese_preview / layer1 / layer2
TV studio	studio_preview / layer1 / layer2 / layer3
Office building	office_building_ex1 / ex2

_layer1.png = shell (build first) · **_layer2.png** = furniture on top (next) · **_preview.png** = the target. Stacking layer1 then layer2 in your head *is* the build order.

Source pack (itch.io)	Covers (folder)
Modern Interiors — limezu.itch.io/moderninteriors	base furniture, the Singles system
Modern Exteriors — limezu.itch.io/modernexteriors	streets, buildings, vehicles → exteriors/
Modern Office — limezu.itch.io/modernoffice	HQ desks/chairs → Modern Office sheets
Modern User Interface — limezu.itch.io/modernuserinterface	UI / HUD art → ui/
Modern Farm — limezu.itch.io/modernfarm	crops, animals, fences → farm/

You are the namer. LimeZu never labels `Singles_42`. Whatever you type becomes the official name. Pick clear, consistent names: `gym_treadmill`, not `thing3`.

A Clean Day's Workflow

1. **Start the server** — `npm start`, open `localhost:8080/`.
2. **Pick a target room** from `examples/`; open its `_preview.png` beside the browser.
3. **Name any missing pieces (Job A)** — LimeZu → sprite-browser; Office → cutter+labeler; new tile → tile-inspector.
4. **QA the cuts** — `verify-sprites.html`, scroll, fix anything cut off or doubled.
5. **Build the room (Job B)** — press **E**. Walls tab: outline + doorways + boundaries. Furniture tab: place to match the preview. Use Show Collisions to check paths.
6. **Save** — **X** on each tab.
7. **Hand off the files** to the engineer.

When Something Looks Wrong

You see…	Means…	Fix
Item isn't in the editor list	Never named	Name it (Job A before Job B)
Stripes / smear	Cut at wrong grid size	Re-cut; for LimeZu use the Single, never a theme crop
Magenta squares on floor/walls	Wrong tile frame #	tile-inspector.html → copy correct frame
Only half an object	Multi-tile placed as one	Use the bigger Single, or place matching tiles together
Character walks through a wall	Missing boundary	Walls tab → add a wall there; Show Collisions to confirm